

PH 607: Physics of Ultracold Quantum Gases

Course Code	PH 607
Course Name	Physics of Ultracold Quantum Gases
L-T-P-C	3-0-0-3
Prerequisites	PH301/PH513 (Quantum Mechanics), PH522 (Statistical Mechanics), PH524/EP403(Physics of Atoms and Molecules)
Intended for	UG/PG/I-PhD/PhD elective
Mutual Exclusion	None
Approval	46th BoA (May 2022)

Course Contents

- **The ideal Bose gas:** The Bose Einstein condensation in ideal Bose gases, Off-diagonal long-range order, Transition temperature and condensate fraction, velocity distribution, Thermodynamic quantities. [6 Hours]
- **Manipulation of atomic internal and external degrees of freedom:** Level structure and atomic transitions of alkali-metal atoms, Atom-field interaction, Cooling, trapping and imaging ultracold gases. [6 Hours]
- **Atom-atom interaction:** Contact interaction, scattering length, Feshbach resonances, Dipolar long-range interactions. [4 Hours]
- **Bose-Einstein condensates (BEC):** Condensation and Gross-Pitaevskii equation (GPE) for the macroscopic wave function, BEC dynamics in uniform and trapped configurations; Thomas-Fermi approximation, Hydrodynamic equations, Elementary excitations, BEC as simulator of quantum vacuum effects (Hawking radiation and Casimir effect), collapse and supersolidity with dipolar quantum gases. [12 Hours]
- **Atomic mixtures:** Coupled GPE, spin waves, phase diagram, Josephson effect and magnetism, Quantum droplets. [7 Hours]
- **Lower dimensional systems as solid-state quantum simulators:** Phase fluctuations, Mermin-Wagner-Hohenberg theorem, optical lattices, Bose-Hubbard model, Entanglement and correlations. [7 Hours]

Text Books

1. C.J. Foot, **Atomic Physics**, Oxford University Press, 2005.
2. C.J. Pethick and H. Smith, **Bose-Einstein condensation in dilute gases**, Cambridge University Press, 2008.

References

1. M. Ueda, **Fundamentals and New Frontiers of Bose-Einstein Condensation**, World Scientific Publishing Company 2010.
2. Lev Pitaevskii and Sandro Stringari, **Bose-Einstein Condensation, and Superfluidity**, Oxford Science Publication, 2016
3. F. Dalfovo, S. Giorgini, Lev P. Pitaevskii, and S. Stringari, **Theory of Bose-Einstein condensation in trapped gases**, Rev. Mod. Phys. 71, 463 (1999)